

Reflective Journal: Module 3 Machine Learning Workflow

This lab helped me understand machine learning as more of a full process instead of just one step where a model gives an answer. Before this assignment, I mostly thought of machine learning as training a model and checking if it was right or wrong. After working through the Wine classification project, I can see that the model is only one part of the workflow. The data has to be loaded, explored, split, trained on, tested, and then evaluated carefully. That made machine learning feel more organized, but also more detailed than I first expected.

The main focus of this lab was using the Wine dataset to classify different wine classes based on numerical features like alcohol, malic acid, ash, color intensity, and proline. The dataset had 178 samples, 13 features, and three target classes. One thing that stood out to me early was that the dataset was clean, with no missing values. That made the lab easier to follow, but it also made me realize that real-world data probably will not always be this neat. In a real situation, checking for missing values and class balance would probably be a bigger part of the work.

One of the biggest things that clicked for me was the difference between supervised, unsupervised, and reinforcement learning. Supervised learning made the most sense in this lab because the dataset already had labels for each wine class. It was like the model had an answer key during training. Unsupervised learning is different because the model would be looking for patterns or groups without being told the correct answers ahead of time. Reinforcement learning also became clearer when I thought about it as trial and error, like a system learning from rewards and penalties. The examples in the assessment helped me understand that the type of machine learning depends on what kind of data is available and what the goal is.

The train/test split was another important part of the lab for me. At first, splitting the data seemed like just a setup step, but I understand now why it matters. The training set is what the model learns from, while the test set checks whether the model can handle data it has not already seen. That helped me understand the difference between memorizing and actually learning patterns. If a model only performs well on the data it trained on, that does not really prove it is useful. Testing it on unseen data gives a better idea of whether it can generalize.

Comparing Logistic Regression and the Decision Tree was probably the most interesting part. Logistic Regression had the better accuracy at 88.9%, while the Decision Tree had 83.3%. Before this lab, I probably would have assumed the Decision Tree might do

better because it seems more flexible and easier to visualize. But this showed me that the “better” model depends on the data and how well the model handles the problem. A model that sounds more advanced or more flexible is not automatically better.

The confusion matrix also helped me slow down and think beyond accuracy. At first, I wanted to just look at the accuracy score and call that the result. But the confusion matrix showed where the model was actually making mistakes. Logistic Regression classified class_0 very well, but it had some confusion between class_1 and class_2. That made evaluation feel more realistic. Accuracy gives the overall score, but it does not explain the specific weaknesses of the model. I can see now why people use multiple evaluation tools instead of relying on one number.

The feature experiment was another good learning moment. I chose alcohol, color intensity, and proline as my three features. My model reached 83.3% accuracy, which was lower than the original Logistic Regression model. At first, that felt like I had picked the “wrong” features, but I think that was actually the point of the exercise. It showed me that feature selection matters a lot. Some features may seem useful, but the model’s performance depends on how well those features actually separate the classes. It also made me wonder how much better the model would perform if I tested more combinations or used all features with scaling.

One challenge I had was keeping all the steps connected in my head. The code itself was not impossible to follow, but there were a lot of moving parts: selecting features, splitting the data, training two models, calculating accuracy, reading the classification report, and then interpreting the confusion matrix. I had to remind myself that these steps are not random. They are part of a workflow. Each step answers a different question. What data do we have? What features are we using? How will we test the model? How well did it perform? Where did it make mistakes?

This lab also helped me connect machine learning to real-world applications. For example, in healthcare, supervised learning could be used to classify patients into risk categories based on lab results, symptoms, or medical history. That connection made the lab feel more useful because the same basic workflow could apply outside of the Wine dataset. The dataset changes, but the process is similar: collect labeled data, prepare it, train a model, test it, and evaluate the results carefully.

If I were improving this project, I would try using StandardScaler because some features have very different ranges. For example, proline has much larger values than features like alcohol or ash. Since Logistic Regression can be affected by feature scaling, standardizing the data might improve performance. I would also experiment with

different feature combinations instead of choosing only one set. Another thing I would want to try is using more models, like k-nearest neighbors or random forest, to see how they compare.

Overall, this lab made the machine learning workflow feel much more real to me. I learned that building a model is not just pressing run and getting an accuracy score. It involves understanding the data, making choices about features, testing fairly, and interpreting the results. The biggest takeaway for me is that machine learning is not only about the final answer. It is about the process of figuring out whether the answer is reliable, where the model struggles, and what could be improved next.